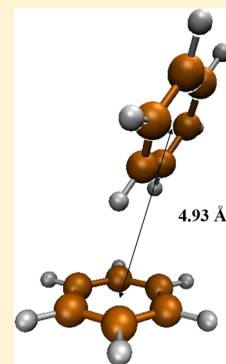


Chemical Dynamics Simulations of Benzene Dimer Dissociation

Xinyou Ma, Amit K. Paul, and William L. Hase*

Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, Texas 79409, United States

ABSTRACT: Classical chemical dynamics simulations were performed to study the intramolecular and unimolecular dissociation dynamics of the benzene dimer, $Bz_2 \rightarrow 2 Bz$. The dissociation of microcanonical ensembles of Bz_2 vibrational states, at energies E corresponding to temperatures T of 700–1500 K, were simulated. For the large Bz_2 energies and large number of Bz_2 vibrational degrees of freedom, s , the classical microcanonical (RRKM) and canonical (TST) rate constant expressions become identical. The dissociation rate constant for each T is determined from the initial rate $dN(t)/dt$ of Bz_2 dissociation, and the $k(T)$ are well-represented by the Arrhenius eq $k(T) = A \exp(-E_a/RT)$. The E_a of 2.02 kcal/mol agrees well with the Bz_2 dissociation energy of 2.32 kcal/mol, and the A-factor of $2.43 \times 10^{12} \text{ s}^{-1}$ is of the expected order-of-magnitude. The form of $N(t)$ is nonexponential, resulting from weak coupling between the Bz_2 intramolecular and intermolecular modes. With this weak coupling, large Bz_2 vibrational excitation, and low Bz_2 dissociation energy, most of the trajectories dissociate directly. Simulations, with only the Bz_2 intramolecular modes excited at 1000 K, were also performed to study intramolecular vibrational energy redistribution (IVR) between the intramolecular and intermolecular modes. Because of restricted IVR, the initial dissociation is quite slow, but $N(t)$ ultimately becomes exponential, suggesting an IVR time of 20.7 ps.



I. INTRODUCTION

Association of polycyclic aromatic hydrocarbon (PAH) molecules has been suggested as an important initiation step in soot formation during combustion.^{1–4} The dominant molecular species, extracted from small carbon particles present in flames, are as small as three- to four-ring aromatics and extend to sizes larger than coronene.^{5–10} Compared to the energies associated with high-temperature combustion experiments, the binding energies of these relatively small aromatic molecules are quite low^{11,12} and the initiation steps for soot formation from such PAH molecules remain uncertain.

Schuetz and Frenklach performed chemical dynamics simulations to study the collisional association of pyrene molecules¹³ and lifetimes of the resulting pyrene dimers. This work was followed by a related simulation by Wong et al.,¹⁴ in which lifetimes were determined of dimers formed by the association of molecules consisting of aromatics linked by aliphatic chains. From this work, it was proposed that lifetimes of the dimers are sufficiently long that they can survive to form carbon nuclei at high-temperature flame conditions. However, this suggestion, reached from the molecular dynamics simulations, is not consistent with later studies. Combined experiments and calculations investigated the possibility of pyrene dimerization and nucleation at high temperatures,¹⁵ and they indicated that pyrene dimerization cannot be an important step in soot formation. Pyrene dimer was not observed experimentally, and the calculations found that equilibrium for dimerization favors dissociation of the weakly bound pyrene complex. This conclusion is supported by a molecular dynamics simulation¹⁶ for $2 \text{ pyrene} \rightleftharpoons \text{pyrene}_2$, which found that equilibrium favors dimer dissociation at high temperatures. A consideration of thermodynamic properties for PAH molecules and their dimers (i.e., their small binding energies and large

entropy decreases upon dimer formation), indicates that PAH_2 dimer formation is unimportant for moderate-sized PAH molecules at flame temperature.¹²

A possible important dynamical property of PAH – PAH complexes is weak coupling between the intramolecular vibrational modes of the PAH molecules and the much lower frequency intermolecular vibrational modes formed by the $PAH + PAH$ association. Such dynamics are present in and have been studied for van der Waals molecules^{17–19} and ion–molecule complexes,^{20–22} giving rise to unimolecular dissociation dynamics different than that predicted by the RRKM theory of unimolecular dynamics,²³ which assumes rapid intramolecular vibrational redistribution (IVR) between all the vibrational modes of the unimolecular reactant.^{24–26} IVR may be inefficient for PAH_2 dimers, and vibrational states with excited intramolecular modes may be particularly long-lived as a result of slow energy transfer from the intramolecular to intermolecular modes. The latter modes are directly coupled to the $PAH_2 \rightarrow 2 PAH$ dissociation reaction coordinate. As a result, neither a fixed energy microcanonical nor a fixed temperature canonical ensemble of PAH_2 dimers will dissociate with an exponential time-dependence as assumed by RRKM theory.^{24,26,27}

In the work presented here, chemical dynamics simulations are performed for the unimolecular dissociation of the benzene dimer (Bz_2) to investigate the possibility of slow IVR and non-RRKM dynamics for aromatic dimers and clusters. The Bz_2 dimer is randomly vibrationally excited at fixed energies and temperatures, to determine if its time-dependent dissociation

Received: April 23, 2015

Revised: May 28, 2015

probability is exponential as predicted by RRKM theory. Simulations are also performed, with only the intramolecular modes of the Bz molecules excited, to study the efficiency of energy transfer between the intermolecular and intramolecular modes. Of interest is to determine if there are intramolecular vibrational states of Bz₂ with lifetimes much longer than that of RRKM theory.

II. COMPUTATIONAL PROCEDURE

A. Potential Energy Function. The analytic potential energy function for the benzene dimer was written as the sum

$$V = V_{\text{benzene}} + V_{\text{benzene,benzene}} \quad (1)$$

where V_{benzene} is the benzene intramolecular potential and $V_{\text{benzene,benzene}}$ is the benzene–benzene (Bz₂) intermolecular potential. The latter is represented by the OPLS aromatic–aromatic force field.²⁸ This model gives the “T-tilted” geometry reported by Jorgensen et al.²⁸ for the benzene global potential energy minimum (see Figure 1), which is in good agreement

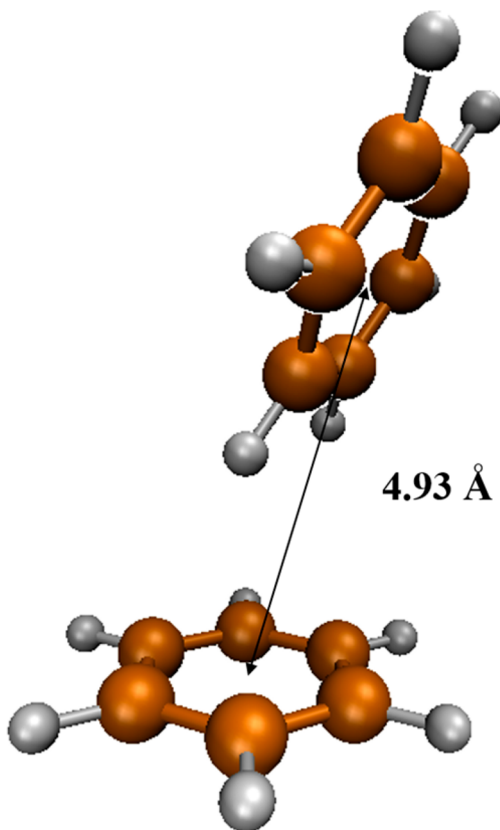


Figure 1. Minimum energy geometry for the benzene dimer using the OPLS intermolecular potential parameters.²⁸

with the geometry found from CCSD(T)/CBS calculations.²⁹ The OPLS benzene–benzene center-of-mass separation is 4.93 Å compared to the ab initio value of 4.96 Å. The OPLS energy for the potential energy minimum is −2.32 kcal/mol, compared to the CCSD(T)/CBS value of −2.84 kcal/mol.²⁹ The six intermolecular frequencies for the benzene dimer, as given by OPLS, are 6.4, 7.7, 24, 37, 41, and 65 cm^{−1}. For comparison, the frequencies obtained from a parametrization of the nonempirical model (NEMO) empirical intermolecular potential, to CCSD(T) calculations for the benzene dimer, are 3.4, 4.8, 13.6, 38.2, 42.1, and 53.7 cm^{−1}.³⁰ An “average”

representative intermolecular potential energy curve for the benzene dimer is obtained by averaging the potential energies of random orientations of the benzene molecules versus the benzene–benzene center-of-mass separation. This curve is shown in Figure 2. Its minimum potential energy is −0.54 kcal/mol at a center-of-mass separation of 6.7 Å.

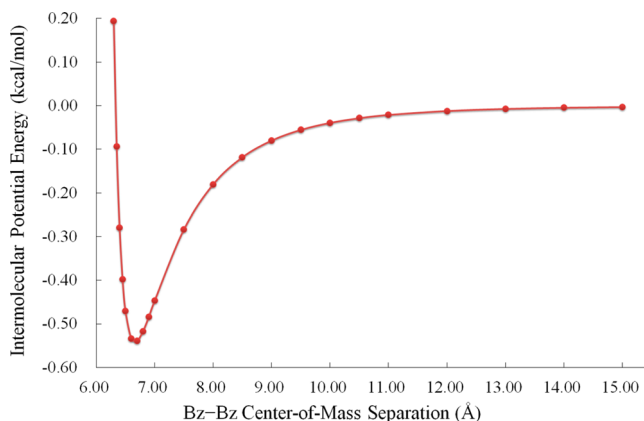


Figure 2. Orientation averaged benzene–benzene (Bz–Bz) intermolecular potential versus the Bz–Bz center-of-mass separation.

The benzene intramolecular potential was represented by a valence force field consisting of in-plane C–H and C–C stretching, H–C–C and C–C–C bending force constants, out-of-plane H–C wagging force constants, and C–C–C–C torsional potentials. Force fields for benzene have been extensively studied through both ab initio computations^{31–33} and experiments.^{34–37} The force field used here was developed from the Goodman, Ozkabak, and Thakur (GOT) experimental “benchmark” force field for benzene.³⁷ Except for one small change described below, the in-plane force constants are the same as those given by GOT. Instead of using a quadratic term for the C–C–C–C torsional potential, it was deemed more appropriate to represent it by the function

$$V(\phi) = V_0(1 - \cos 2\phi)/2 \quad (2)$$

for which the quadratic ϕ force constant is given by $2V_0 \cos 2\phi$, evaluated at $\phi = 0$.

With eq 2, it was not straightforward to include the torsion–torsion and torsion–wag nondiagonal force constants into the potential. Thus, to accurately represent the out-of-plane frequencies for benzene, the GOT H–C wag diagonal, H–C wag–wag nondiagonal, and C–C–C–C diagonal torsion force constants were varied to obtain the best fit to the GOT harmonic vibrational frequencies. In performing the fitting, it was recognized that more accurate in-plane frequencies were obtained by modifying the C–C–C diagonal and nondiagonal bending force constants. The benzene force field for the current study was determined by minimizing the mean percentage deviation (MPD) between the 30 GOT frequencies and those for the current model, in other words,

$$\text{MPD} = \frac{1}{30} \sum_{i=1}^{30} \left(\frac{\omega_{\text{cur},i} - \omega_{\text{GOT},i}}{\omega_{\text{GOT},i}} \times 100\% \right) \quad (3)$$

where ω_{cur} are the current frequencies. As described above, the force constants that were varied by this fitting are the out-of-plane H–C diagonal and nondiagonal wagging and the diagonal C–C–C–C torsion and in-plane C–C–C diagonal

and nondiagonal bending force constants. The MPD, as compared to the GOT frequencies, is quite small and 1.22%. The force constants for the current potential are listed in Table 1, and the comparison between the GOT and current vibrational frequencies is given in Table 2.

Table 1. Force Constants for the Benzene Intramolecular Potential^{a,b,c,d}

type	term	current	term	GOT ³⁶
$t_i t_i$	f_R	6.6160	f_R	6.616
$s_i s_i$	f_r	5.5470	f_r	5.547
$\alpha_i \alpha_i$	f_θ	1.0780	f_θ	1.284, 1.257
$\phi_i \phi_i$	f_φ	0.5260	f_β	1.047
$\gamma_i \gamma_i$	f_α	0.3120	f_α	0.447
$t_i t_{i+1}$	$f_{RR'(o)}$	0.7280	$f_{RR'(o)}$	0.728
$t_i t_{i+2}$	$f_{RR'(m)}$	-0.4190	$f_{RR'(m)}$	-0.419
$t_i t_{i+3}$	$f_{RR'(p)}$	0.3830	$f_{RR'(p)}$	0.383
$t_i s_i = t_i s_{i+1}$	$f_{rR(o)}$	0.2000	$f_{rR(o)}$	0.200
$t_i s_{i+2} = t_i s_{i+5}$	$f_{rR(m)}$	-0.0080	$f_{rR(m)}$	0.008
$t_i s_{i+3} = t_i s_{i+4}$	$f_{rR(p)}$	-0.1300	$f_{rR(p)}$	-0.130
$s_i s_{i+1}$	$f_{rr'(o)}$	0.0070	$f_{rr'(o)}$	0.007
$s_i s_{i+2}$	$f_{rr'(m)}$	0.0080	$f_{rr'(m)}$	0.008
$s_i s_{i+3}$	$f_{rr'(p)}$	-0.0220	$f_{rr'(p)}$	-0.022
$t_i \alpha_i$	$f_{R\theta(o)}$	0.1240	$f_{R\theta(o)}$	0.124
$s_i \alpha_i$	$f_{r\theta(o)}$	-0.1220	$f_{r\theta(o)}$	-0.135, -0.110
$t_i \phi_i = -t_i \phi_{i+1}$	$f_{R\varphi(o)}$	0.1080	$f_{R\beta(o)}$	0.216
$t_i \phi_{i+2} = -t_i \phi_{i+5}$	$f_{R\varphi(m)}$	-0.0110	$f_{R\beta(m)}$	-0.022
$t_i \phi_{i+3} = -t_i \phi_{i+4}$	$f_{R\varphi(p)}$	0.0160	$f_{R\beta(p)}$	0.033
$s_i \phi_{i+1} = -s_i \phi_{i+5}$	$f_{r\varphi(o)}$	-0.0340	$f_{r\beta(o)}$	-0.068
$s_i \phi_{i+2} = -s_i \phi_{i+4}$	$f_{r\varphi(m)}$	-0.0140	$f_{r\beta(m)}$	-0.027
$\alpha_i \alpha_{i+1}$	$f_{\theta\theta'(o)}$	0.0160		
$\phi_i \phi_{i+1}$	$f_{\varphi\varphi'(o)}$	0.0040	$f_{\beta\beta'(o)}$	0.016
$\phi_i \phi_{i+2}$	$f_{\varphi\varphi'(m)}$	-0.00575	$f_{\beta\beta'(m)}$	-0.023
$\phi_i \phi_{i+3}$	$f_{\varphi\varphi'(p)}$	-0.0005	$f_{\beta\beta'(p)}$	-0.002
$\alpha_i \phi_{i+1} = \alpha_i \phi_{i+5}$	$f_{\theta\varphi(o)}$	-0.0530	$f_{\theta\beta(o)}$	-0.106
	$V_{torsion}$	23.5000		

^aThe CC stretch, CH stretch, CCC angle bend, CCH rocking, and CH wag motions are denoted, respectively, by R , r , θ , φ (or β), and α for the force constant terms and t , s , α , ϕ , and γ for the motion types. The benzene torsion motion is denoted as $V_{torsion}$ for the force constant term. ^bUnits for the stretch, bend, wag, torsion, stretch–stretch, stretch–bend, bend–bend force constants are mdyn/Å, mdyn·Å/rad², mdyn·Å/rad², kcal/mol, mdyn/Å, mdyn/rad, and mdyn·Å/rad², respectively. ^cThe C–H and C–C equilibrium distances are taken as 1.084 and 1.397 Å, respectively. ^dThe CCH rocking angle, β , used for the GOT force field is defined as $\beta = (\varphi - \varphi')/2$.³⁷ This definition is used to determine the GOT φ force constants for the CCH rocking angle.

B. Chemical Dynamics Simulations. The chemical dynamics simulations of the unimolecular dissociation of the benzene dimer (Bz_2) were performed with the computer program VENUS.^{38,39} The simulations were performed to model dissociation of randomly excited Bz_2 at temperatures of 700, 800, 900, 1000, 1200, and 1500 K. Representative quantum mechanical energies E for these temperatures were determined using the harmonic oscillator model⁴⁰ for the sixty-six vibrational modes of Bz_2 and the values of E for the above respective temperatures are 156.1, 164.3, 173.1, 182.4, 202.2, and 234.3 kcal/mol. These are total energies and include the 127.1 kcal/mol harmonic zero-point energy (ZPE) for Bz_2 .

Quantum microcanonical normal mode sampling⁴¹ was used to add these energies to Bz_2 about its “T-tilted” potential

Table 2. Comparison of Current and GOT Benzene Harmonic Vibrational Frequencies^a

sym.	mode assignment		vibrational frequency (cm ⁻¹)	
	Wilson no.	Herzberg no.	current	GOT
e _{2u}	ω_{16}	ω_{20}	399.03	398
			399.03	398
e _{2g}	ω_6	ω_{18}	612.85	607.8
			612.87	607.8
a _{2u}	ω_{11}	ω_4	653.46	674
b _{2g}	ω_4	ω_8	696.18	707
e _{1g}	ω_{10}	ω_{11}	846.22	847.1
			846.23	847.1
e _{2u}	ω_{17}	ω_{19}	1011.70	967
			1011.70	967
b _{2g}	ω_5	ω_7	987.61	990
a _{1g}	ω_1	ω_2	1022.90	994.4
b _{1u}	ω_{12}	ω_6	1090.10	1010
e _{1u}	ω_{18}	ω_{14}	1030.70	1038.3
			1030.80	1038.3
b _{2u}	ω_{15}	ω_{10}	1097.80	1149.7
e _{2g}	ω_9	ω_{17}	1154.00	1177.8
			1154.00	1177.8
b _{2u}	ω_{14}	ω_9	1374.50	1309.4
a _{2g}	ω_3	ω_3	1387.80	1367
e _{1u}	ω_{19}	ω_{13}	1567.40	1494
			1567.40	1494
e _{2g}	ω_8	ω_{16}	1708.40	1607
			1708.40	1607
e _{2g}	ω_7	ω_{15}	3174.40	3174
			3174.40	3174
b _{1u}	ω_{13}	ω_5	3189.50	3174
e _{1u}	ω_{20}	ω_{12}	3179.50	3181.1
			3179.50	3181.1
a _{1g}	ω_2	ω_1	3192.50	3191

^aThe GOT frequencies are in ref 37.

energy minimum. An ensemble of 1000 trajectories were calculated for each of the above energies, and each trajectory was integrated for 50 ps, unless $Bz_2 \rightarrow Bz + Bz$ dissociation occurred. Tests showed that once the Bz molecules acquired a center-of-mass separation of 14 Å, dissociation occurred and the system did not return to the Bz_2 potential energy minimum. This separation was used to identify the time for dissociation. The average Bz – Bz potential energy (Figure 2) for this separation is -0.0045 kcal/mol.

For the above quantum microcanonical sampling, both the intramolecular and intermolecular vibrational modes of Bz_2 are sampled randomly. To compare with the results of these simulations, a simulation was performed in which only the 60 intramolecular modes of Bz_2 were sampled by quantum microcanonical sampling. This simulation was performed for $E = 182.4$ kcal/mol (i.e., $T = 1000$ K). ZPE was added to the six intermolecular modes of Bz_2 for this simulation.

C. Analyses of the Trajectory Results. The quasiclassical procedure, used to transform the initial conditions for the quantum microcanonical ensemble to Cartesian coordinates and momenta, chooses a random phase for each normal mode to partition the mode's energy to potential and kinetic.⁴⁰ However, the six intermolecular modes of the Bz – Bz dimer are highly anharmonic, and large potential energies are not correctly added to these modes by this method. Thus, to add the mode energy, for an initial condition chosen with a large

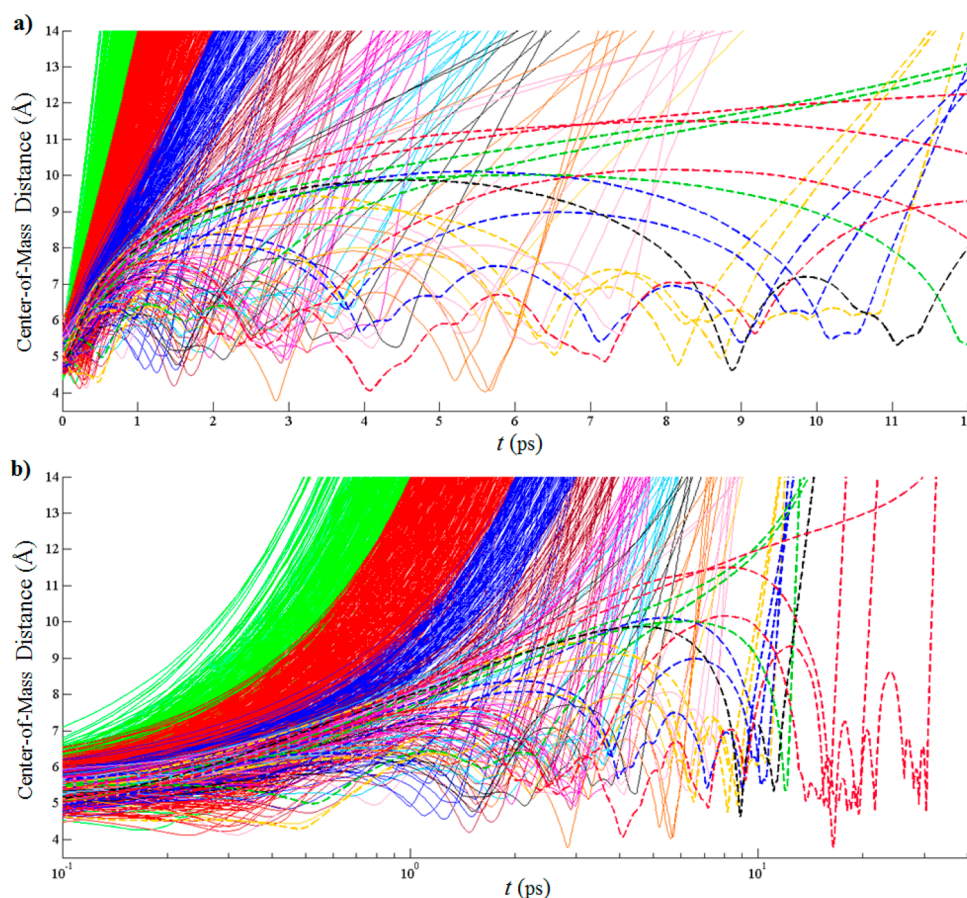


Figure 3. Plots of Bz–Bz center-of-mass separation versus time, for dissociating trajectories at 1000 K. The trajectories are color-coded for dissociation in 1 ps intervals: 0–1, green; 1–2, red; 2–3, dark blue; 3–4, brown; 4–5, purple; 5–6, light blue; 6–7, black; 7–8, orange; 8–9, pink; 9–10, gold; 10–11, gold; 11–12, blue; 12–13, dark blue; 13–14, green; 14–15, black; >15, red. The trajectories dissociating before and after 10 ps are plotted in solid and broken lines, respectively. In (a), the time along the x axis is linear, and in (b), it is logarithmic.

energy in an intermolecular mode, all the energy is added as kinetic. This maintains the proper internal energy distribution for the quantum microcanonical ensemble, but initial conditions are not selected with a large Bz–Bz separation and there is a time lag t_0 for $\text{Bz}_2 \rightarrow \text{Bz} + \text{Bz}$ dissociation. As shown in previous work,^{42–44} if the motions of the unimolecular reactant's vibrational modes are highly coupled, there is no time lag for the unimolecular reaction, and the initial $t = 0$ rate constant is that for the quantum microcanonical ensemble. However, as shown for the following simulation results, there is weak coupling between the Bz_2 intermolecular and intramolecular modes, and there is a time lag t_0 for $\text{Bz}_2 \rightarrow \text{Bz} + \text{Bz}$ dissociation.

For each constant energy simulation, the number of benzene dimers remaining undissociated at time t was fit to the biexponential function

$$\frac{N(t - t_0)}{N(t_0)} = f_1 e^{-k_1(t-t_0)} + f_2 e^{-k_2(t-t_0)} \quad (4)$$

where t_0 is the time lag for dissociation and taken as the time the first benzene dimer dissociates, $N(t_0)$ is the total number of benzene dimers at t_0 (i.e., 999), $f_1 + f_2 = 1$, and k_1 and k_2 are unimolecular rate constants. The unimolecular rate constant at $t = t_0$ is found by taking the derivative $d[N(t - t_0)/N(t_0)]/dt$ and taking the limit $t \rightarrow t_0$, in other words,

$$k(t_0) = f_1 k_1 + f_2 k_2 \quad (5)$$

These $k(t_0)$ are taken as the unimolecular rate constants for the initial quantum microcanonical ensembles. The biexponential in eq 4 was used for it provides a much better fit to the simulation results than does a single exponential. No physical meaning is assigned to the fitting constants k_1 and k_2 .

The energy for each simulation is the average energy for temperature T , and the $k(t_0)$ for these temperatures were fit to the Arrhenius equation $k = A \exp[-(E_a/RT)]$ to obtain the A-factor and activation energy for $\text{Bz}_2 \rightarrow \text{Bz} + \text{Bz}$ dissociation.

III. SIMULATION RESULTS AND DISCUSSION

A. Random Excitation of the Bz_2 Intermolecular and Intramolecular Modes. 1. *Nature of the Dissociating Trajectories.* With their very low vibrational frequencies (section A), the energy distributions are classical for the intermolecular modes in the ensembles of randomly excited Bz_2 dimers. The average energy in an intermolecular mode at temperature T is $(E_{\text{ZPE}} + RT)$, where E_{ZPE} is the mode's zero-point energy (ZPE) which is quite small. At 1000 K, the average energy of the 65 cm^{-1} mode is 2.08 kcal/mol, which includes a harmonic ZPE of 0.09 kcal/mol.

For the OPLS intermolecular potential used for Bz_2 , the $\text{Bz}_2 \rightarrow \text{Bz} + \text{Bz}$ dissociation energy is 2.32 kcal/mol. Thus, for the 700–1500 K simulations, there are many trajectory initial conditions for which some or all of the intermolecular modes each have an initial energy in excess of the Bz_2 dissociation energy. As a result of these “large” intermolecular energies,

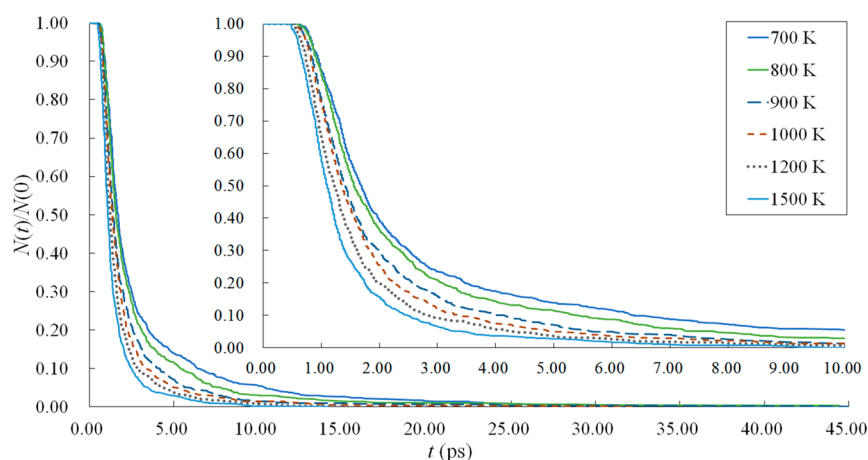


Figure 4. Ratio of undissociated Bz₂ dimers, $N(t)$, and the initial number of Bz₂ dimers, $N(0)$, versus time for the different simulation temperatures.

many of the trajectories dissociate directly without random IVR between the Bz₂ intermolecular and intramolecular modes. This is illustrated in Figure 3 for the 1000 K simulations, where the Bz–Bz center-of-mass separation is plotted versus time for each dissociating trajectory. The trajectories are color-coded for dissociation in 1 ps intervals. Trajectories which dissociate directly do not have an outer turning point in the Bz–Bz center-of-mass motion. Within statistical uncertainties, one-half of the trajectory initial conditions have a negative and one-half have a positive Bz–Bz center-of-mass momentum. Trajectories for the former, which dissociate directly, have an inner turning point.

As shown in Figure 3, all of the reaction within the first 2 ps is direct, and the vast majority of the reaction is direct for the 2–3 ps. For the later dissociations, oscillations in the Bz–Bz center-of-mass separation become important before dissociation occurs. As illustrated in more detail in the next section, the percentage of the dissociation which occurs within the first 3 ps is 76.5, 79.1, 84.1, 87.9, 90.8, and 93.4% for the temperatures of 700, 800, 900, 1000, 1200, and 1500 K, respectively. Thus, the vast majority of the dissociations is direct. The importance of direct dissociation increases with an increase in temperature. The respective percentage of the dissociation, which is direct, is 84.9, 95.2, and 97.2% for T of 700, 1000, and 1500 K.

As discussed above, the importance of direct dissociation is related to trajectory initial conditions with the intermolecular modes sufficiently excited to directly dissociate. The probability that the 65 cm^{−1} Bz–Bz stretching reaction coordinate initially contains energy in excess of the $E_0 = 2.32$ kcal/mol dissociation energy is²³

$$P(E \geq E_0) = \int_{E_0 - \text{ZPE}}^{\infty} e^{-E/k_B T} dE/k_B T \quad (6)$$

where ZPE is the mode's zero-point energy. For 1000 K, this probability is 0.329. The probability that the six intermolecular modes of the Bz₂ dimer initially contain energy in excess of E_0 is

$$P(E \geq E_0) = \int_{E_0 - \text{ZPE}}^{\infty} E^5 e^{-E/k_B T} dE / [5! (k_B T)^6] \quad (7)$$

where ZPE is the total zero-point energy of the six intermolecular modes. For 1000 K, this probability is 0.999. As discussed above, 0.952 of the dissociation is direct at 1000 K. The probability that the three Bz₂ intermolecular mode of 65, 41, and 37 cm^{−1} contain energy in excess of E_0 is 0.929,

which is nearly identical to the direct dissociation probability. Similar agreement is also obtained at 700 and 1500 K, where the direct dissociation probabilities are 0.849 and 0.972, respectively, as discussed above, whereas the probabilities that those three modes contain energy in excess of E_0 are 0.843 and 0.973, respectively. These probabilities may also be found by a “census” of the initial conditions, which gives statistically the same results as given here.

2. $N(t)$, Number of Bz₂ versus Time. The ratio of undissociated Bz₂ dimers, $N(t)$, and the initial number of Bz₂ dimers, $N(0)$, is plotted versus time in Figure 4 for the different simulation temperatures with random excitation of both the Bz₂ intramolecular and intermolecular modes. As illustrated in Figure 4, and discussed above in Section C, there is a time lag t_0 for Bz₂ → Bz + Bz dissociation. The value of t_0 for each simulation temperature was taken as the time the first dissociation occurred. The resulting t_0 varied from 0.58 ps at 700 K to 0.42 ps at 1500 K. The distributions $N(t - t_0)/N(t_0)$ were fit by the biexponential function in eq 4 and excellent fits were obtained. Representative fits are illustrated in Figure 5 as plots of $\ln[N(t - t_0)/N(t_0)]$ versus t , for T of 700 and 1200 K. The parameters for the biexponential fits are given in Table 3.

The largest exponential component in the fits is the one with the largest rate constant, and its fraction f_1 is ~ 0.8 or larger. The k_1/k_2 rate constant ratios are similar and range from 4.2 to 5.1.

3. $k(T)$ and the Arrhenius Parameters. The unimolecular rate constant at t_0 [i.e., $k(t_0)$] is assumed to represent that for the initial quantum microcanonical ensemble. Equation 5 gives the relationship between $k(t_0)$ and the fitting parameters for the $N(t - t_0)/N(t_0)$ distribution. The $k(t_0)$ are also unimolecular rate constants $k(T)$ for the different simulation temperatures, and in Figure 6 $\ln k(t_0)$ is plotted versus $1/T$ and fit to the Arrhenius equation $\ln k = \ln A - E_a/RT$. The simulation $\ln k(t_0)$ versus $1/T$ plot is well-represented by this linear function, and the Arrhenius parameters are $A = 2.43 \times 10^{12} \text{ s}^{-1}$ and $E_a = 2.02$ kcal/mol. The value for E_a is similar to the Bz₂ binding energy of 2.32 kcal/mol. In contrast to this finding for Bz₂ dissociation, the activation energies for dissociation of the Na⁺(Bz) and Na⁺(Bz)₂ clusters do not equal their dissociation energies.⁴⁵ This is a result of temperature-dependent anharmonicity, which is apparently not important for Bz₂ dissociation.

The A-factor is quite small compared to other dissociations (e.g., C–C bond dissociation for alkanes have A-factors which

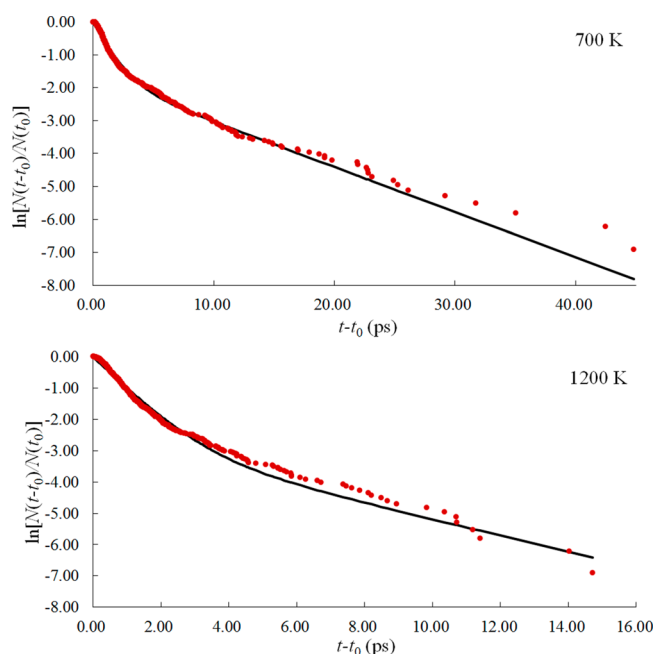


Figure 5. Plots of the simulation $\ln[N(t - t_0)/N(t_0)]$ versus t , and fits by eq 4, for T of 700 and 1200 K.

Table 3. Parameters for Fits to the Simulation $N(t - t_0)/N(t_0)^a$

T (K)	f_1	f_2	k_1	k_2	$k(t_0)$
700	0.795	0.205	0.670	0.147	0.563
800	0.850	0.150	0.780	0.162	0.687
900	0.890	0.110	0.880	0.192	0.804
1000	0.905	0.095	0.950	0.220	0.881
1200	0.925	0.075	1.087	0.260	1.025
1500	0.970	0.030	1.270	0.250	1.239

^a $f_1 + f_2 = 1$, and k_1 , k_2 , and $k(t_0)$ are in units of ps^{-1} .

range from $\sim 10^{15}$ – 10^{17} s^{-1}).^{46–49} To understand the small A-factor for Bz_2 dissociation, consider the transition state theory (TST) expression for a unimolecular dissociation reaction with both intermolecular and intramolecular degrees of freedom, in other words,

$$k(T) = \frac{k_b T}{h} \frac{Q_{\text{inter}}^\ddagger Q_{\text{intra}}^\ddagger}{Q_{\text{inter}} Q_{\text{intra}}} e^{-E_0/k_b T} \quad (8)$$

where the partition functions (Q) in the numerator and denominator are for the TS and dimer, respectively, and E_0 is the potential energy difference between the TS and dimer. Since the intramolecular degrees of freedom are nearly identical for the TS and dimer and one of the intermolecular modes of the dimer is the reaction coordinate, $k(T)$ becomes

$$k(T) \simeq \nu_{\text{rc}} \frac{Q_{\text{inter}}^\ddagger}{Q_{\text{inter}}} e^{-E_0/k_b T} \quad (9)$$

where ν_{rc} is the frequency for the reaction coordinate motion and $Q_{\text{inter}}^\ddagger/Q_{\text{inter}}$ is the ratio of the partition functions for five intermolecular modes of the TS and dimer. Since the ZPE's for the intermolecular modes of the TS and dimer are quite small, the classical and quantum values for E_0 are similar. For classical TST, E_0 is also the Arrhenius activation energy E_a .⁴⁴

The 65 cm^{-1} intermolecular mode is the Bz – Bz center-of-mass stretching motion for the reaction coordinate, which has a harmonic frequency of $1.95 \times 10^{12} \text{ s}^{-1}$. This motion is highly anharmonic, so the actual ν_{rc} frequency will be smaller. The five intermolecular modes, for both the Bz_2 dimer and TS partition functions, are rotational and torsional degrees of freedom.⁵⁰ These modes may be “free rotors” in the quite “loose” TS, but for the energized dimer, these rotations may be somewhat restricted. Thus, the $Q_{\text{inter}}^\ddagger/Q_{\text{inter}}$ ratio is expected to be greater than unity. A value for ν_{rc} less than $2.0 \times 10^{12} \text{ s}^{-1}$ and a $Q_{\text{inter}}^\ddagger/Q_{\text{inter}}$ greater than unity are consistent with the simulation A-factor of $2.43 \times 10^{12} \text{ s}^{-1}$.

B. Random Excitation of the Bz_2 Intramolecular Modes. The above simulations show that, if the Bz_2 dimer is excited randomly, its dissociation probability is nonexponential as a result of inefficient IVR for the dimer's vibrational modes. The plots of the Bz – Bz center-of-mass separation versus time, in Figure 3 for the dissociating trajectories, illustrate inefficient IVR between the intermolecular and intramolecular modes of Bz_2 , indicating this is the origin of the nonexponential dissociation. To study these dynamics in more detail, a simulation was performed in which the Bz_2 intramolecular modes were excited using quantum microcanonical sampling

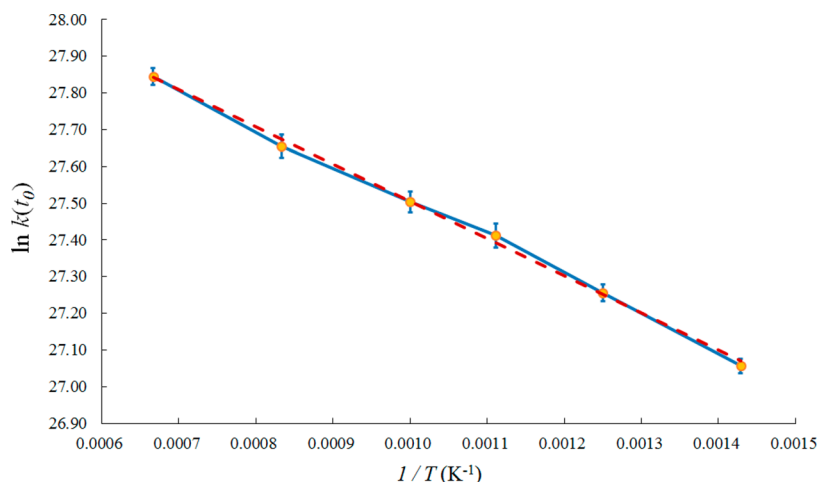


Figure 6. Points: $\ln k(t_0)$, with $k(t_0)$ in sec^{-1} , versus $1/T$ (K^{-1}). Dashed line: fit by the Arrhenius equation. The blue lines connect the points and are not fits.

with 182.4 kcal/mol, for a temperature of 1000 K, and only ZPE was added to the six Bz_2 intermolecular modes.

The results of this 1000 K simulation are shown in Figure 7. Bz_2 dissociation is much slower if only the intramolecular

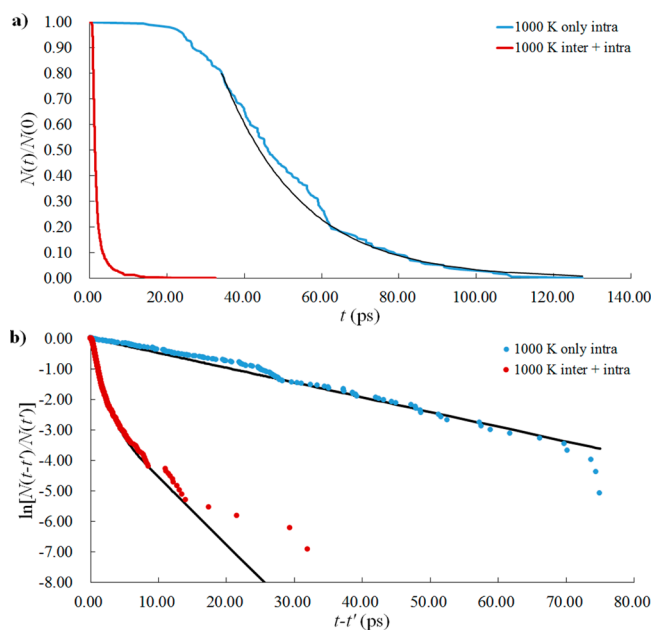


Figure 7. Simulation results for 1000 K. (a) Plots of $N(t)/N(0)$, see Figure 4, for both the intermolecular and intramolecular modes of Bz_2 randomly excited, red curve, and only the intramolecular modes randomly excited blue curve; b) plots of $\ln[N(t - t')/N(t')]$ for the two excitation patterns in (a), with $t' = t_0$ for the red curve and $t' = t_{\text{shift}}$ for the blue curve.

modes are excited (e.g., with both the intermolecular and intramolecular modes randomly excited 90% of the Bz_2 dimers dissociate within 3.36 ps), while if only the intramolecular modes are excited, it takes 78.5 ps for this percentage dissociation. If only 10% dissociation is considered, it takes the intermolecular + intramolecular and only intramolecular simulations 0.82 and 26.7 ps, respectively, to attain this percentage. Clearly dissociation is much slower if only the Bz_2 intramolecular modes are excited.

A more quantitative understanding of the timescale for Bz_2 dissociation, with only the intramolecular modes excited, is obtained by considering the long time component for its $N(t)/N(0)$ distribution. If this distribution is shifted by $t_{\text{shift}} = 34.2$ ps, where $N(t)/N(0)$ equals 0.80 as shown in Figure 7b, the resulting plot of $\ln[N(t - t_{\text{shift}})/N(t_{\text{shift}})]$ is linear with a rate constant of 0.0483 ps^{-1} . This rate constant is substantially smaller than the rate constant $k_2 = 0.220 \text{ ps}^{-1}$ for the long time component in $N(t - t_0)/N(t_0)$ with the both the intermolecular and intramolecular modes of Bz_2 randomly excited.

For this simulation with only the Bz_2 intramolecular modes excited, there is an inflection point in the $N(t)/N(0)$ distribution at approximately t_{shift} of 34.2 ps. Before this time, there is transfer of energy from the Bz_2 intramolecular to intermolecular modes so that $Bz_2 \rightarrow Bz + Bz$ dissociation may occur. After this time, the dissociation becomes exponential and it is possible the rate constant of 0.0483 ps^{-1} for the exponential reflects an IVR time of 20.7 ps.²⁵

IV. SUMMARY

In the discussion, we presented: (1) the classical chemical dynamics simulation results reported here, (2) the accuracy of the classical dynamics as compared to quantum dynamics, (3) the possible importance of aromatic dimers as precursors to soot formation in high temperature combustion, and (4) future additional, related simulations.

The unimolecular and intramolecular dynamics of the randomly excited benzene dimer Bz_2 are studied using fixed energy microcanonical ensembles, which correspond to temperatures of 700 to 1500 K. For molecules with a large number of vibrational degrees of freedom s and E much larger than the dissociation energy, which are the case for $Bz_2 \rightarrow Bz + Bz$, the classical microcanonical (RRKM) and canonical (TST) rate constant expressions become equivalent.⁴⁴ Unimolecular rate constants $k(T)$ were determined for Bz_2 dissociation from the initial dissociations of the microcanonical ensembles. The plot of $\ln k(T)$ versus $1/T$ is linear, giving Arrhenius parameters $E_a = 2.02 \text{ kcal/mol}$ and $A = 2.43 \times 10^{12} \text{ s}^{-1}$. The former is in good agreement with the $Bz_2 \rightarrow Bz + Bz$ dissociation energy of 2.32 kcal/mol and the A-factor is of the order-of-magnitude expected from TST for dissociation of the vibrationally excited and quite loose Bz_2 cluster.

The dissociations of the Bz_2 microcanonical ensembles have biexponential time profiles, $N(t)$, of the number of Bz_2 dimers versus time. These nonstatistical dynamics arise from weak coupling between the 6 intermolecular modes of Bz_2 and the remaining 60 intramolecular modes. For each ensemble most of the trajectories dissociate directly, without a vibration in the Bz – Bz center-of-mass separation, reaction coordinate motion. This results from the weak coupling between the Bz_2 intermolecular and intramolecular modes, the low Bz_2 dissociation energy, and the large excitation of the intermolecular modes for the 700–1500 K simulations.

Energy transfer from the Bz_2 intramolecular to intermolecular modes was studied by a simulation with a quantum microcanonical ensemble representing 1000 K excitation in the intramolecular modes and only ZPE in the intermolecular modes. The initial dissociation of this ensemble was very slow. At ~ 34.2 ps, there is an inflection point in $N(t)$ for the Bz_2 dimers and the ensuing decay of $N(t)$ is exponential with an average $\langle t \rangle = 1/k$ of 20.7 ps. This may represent an IVR time²⁵ for energy transfer from the intramolecular to intermolecular modes.

Previous work indicates that the classical simulations reported here give correct intramolecular and unimolecular dynamics for the Bz_2 dimer. Classical chemical dynamics simulations give accurate IVR times for excited CH^{51-53} and NH^{54} local modes in benzene and *N*-methylacetamide, respectively, in comparison to quantum dynamics. Similar agreement between classical and quantum dynamics is expected for energy transfer from the intramolecular to intermolecular modes of Bz_2 , which is exemplified by the accurate intermolecular to intramolecular energy transfer found in chemical dynamics simulations of $N_2 + C_6F_6$ collisions.⁵⁵ The low-frequency intermolecular modes of Bz_2 are well-described by classical statistics and, once excited, classical dynamics is expected to accurately describe their dissociation. A related chemical system is the weakly coupled intermolecular and intramolecular modes for the $X^- \cdots CH_3Y$ S_N2 ion-dipole complexes. Their unimolecular dissociation to $X^- + CH_3Y$ is well-described by classical chemical dynamics.^{21,22,56–58}

A possible uncertainty regarding the accuracy of classical dynamics for Bz₂ dissociation at long times results from the unphysical flow of ZPE in classical simulations.^{59,60} Though classical dynamics gives an accurate IVR time for a benzene-excited CH local mode,^{51–53} at long times, classical dynamics allows ZPE flow from the high frequency to low frequency modes of benzene.⁵⁹ This should enhance energy transfer to the Bz-Bz intermolecular modes, since the Bz low frequency modes are expected to be more strongly coupled to the Bz-Bz intermolecular modes than the high frequency Bz modes. An algorithm has been proposed for restricting this unphysical ZPE flow,⁶⁰ but it may improperly alter the Bz molecular vibrational motion.⁶¹ For the direct dissociations, which dominate Bz₂ → 2 Bz dissociation, this ZPE flow is expected to be unimportant.

The simulations reported here for Bz₂ indicate that this dimer has intramolecular vibrational states with much longer lifetimes than those for the RRKM statistical model of unimolecular dissociation.²³ It is possible that such long-lived states may participate in soot formation by leading to aggregation of aromatic molecules.^{13,14} However, mechanisms for formation of these states need to be investigated. Association of aromatic molecules is expected to lead to initial excitation of the intermolecular modes.^{15–22} Because of the weak coupling between the intermolecular and intramolecular modes of Bz₂, and presumably of other aromatic dimers, the probability may be very low of forming long-lived intramolecular states by association of aromatic molecules.

There are a number of possible extensions of this work for future studies. It would be of interest to study the unimolecular dissociation of Bz₂ from specific intramolecular vibrational states, to investigate state specificity for its dissociation. Previous work^{13–16} has indicated that the pyrene dimer, pyrene₂, is integral in understanding the possible role of aromatics in soot formation. It seems important to calculate the 2 pyrene → pyrene₂ association rate constant versus temperature. In the absence of collisions, all of the pyrene₂ dimers will ultimately dissociate and it is of interest to determine the relative concentration of these dimers versus time [i.e., $N(t)/N(0)$]. There have been preliminary studies of the dissociation of these dimers, formed by pyrene association.^{13,14} Studies of the unimolecular dissociation of pyrene₂ with an initial microcanonical of vibrational states, as considered here for Bz₂, will be informative as well as studies of the pyrene₂ dissociation from specific intramolecular vibrational states. Finally, it is of interest to extend models of non-RRKM dynamics^{62,63} or develop new models to represent the nonexponential $N(t)/N(0)$ found here for Bz₂ dissociation. A particularly interesting finding is the inflection point in $N(t)/N(0)$ for the dissociation with only the Bz₂ intramolecular modes initially excited.

AUTHOR INFORMATION

Corresponding Author

*E-mail: bill.hase@ttu.edu.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The work presented here is a part of a project supported by the Air Force Office of Scientific Research (AFOSR) under Grant FA9550-12-1-0472. The simulations were performed on the Chemdynm computer cluster of the Hase research group.

REFERENCES

- (1) Palmer, H. B.; Cullis, C. F. The Formation of Carbon from Gases. In *Chemistry and Physics of Carbon*; Marcel Dekker: New York, 1965; Vol. 1, 265–325.
- (2) Haynes, B. S.; Wagner, H. Gg. Soot Formation. *Prog. Energy Combust. Sci.* **1981**, 7, 229–273.
- (3) Graham, S. C. The Modeling of the Growth of Soot Particles during the Pyrolysis and Partial Oxidation of Aromatic Hydrocarbons. *Proc. R. Soc. London, Ser. A* **1981**, 377, 119–145.
- (4) Frenklach, M.; Wang, H. Detailed Modeling of Soot Particle Nucleation and Growth. *Symp. Combust., [Proc.]* **1991**, 23, 1559–1566.
- (5) Öktem, B.; Tolocka, M. P.; Zhao, B.; Wang, H.; Johnston, M. V. Chemical Species Associated with the Early Stage of Soot Growth in a Laminar Premixed Ethylene–Oxygen–Argon Flame. *Combust. Flame* **2005**, 142, 364–373.
- (6) Cain, J. P.; Camacho, J.; Phares, D. J.; Wang, H.; Laskin, A. Evidence of Aliphatics in Nascent Soot Particles in Premixed Ethylene Flames. *Proc. Combust. Inst.* **2011**, 33, 533–540.
- (7) Happold, J.; Grotheer, H.-H.; Aigner, M. Soot Precursors Consisting of Stacked Pericondensed PAHs. In *Combustion Generated Fine Carbonaceous Particles: Proceedings of an International Workshop Held in Villa Orlandi*; Anacapri, May 13–16, 2007; Bockhorn, H., D'Anna, A., Sarofim, A. F., Wang, H., Eds.; KIT Scientific Publishing: Karlsruhe, Germany, 2009; pp 277–288.
- (8) Faccinetto, A.; Desgroux, P.; Ziskind, M.; Therssen, E.; Focsa, C. High-Sensitivity Detection of Polycyclic Aromatic Hydrocarbons Adsorbed onto Soot Particles Using Laser Desorption/Laser Ionization/Time-of-Flight Mass Spectrometry: An Approach to Studying the Soot Inception Process in Low-Pressure Flames. *Combust. Flame* **2011**, 158, 227–239.
- (9) Bouvier, Y.; Mihean, C.; Ziskind, M.; Therssen, E.; Focsa, C.; Pauwels, J. F.; Desgroux, P. Molecular Species Adsorbed on Soot Particles Issued from Low Sooting Methane and Acetylene Laminar Flames: A Laser-Based Experiment. *Proc. Combust. Inst.* **2007**, 31, 841–849.
- (10) Maricq, M. M. An Examination of Soot Composition in Premixed Hydrocarbon Flames via Laser Ablation Particle Mass Spectrometry. *J. Aerosol Sci.* **2009**, 40, 844–857.
- (11) Kolakkandy, S.; Pratihari, S.; Aquino, A. J. A.; Wang, H.; Hase, W. L. Properties of Complexes Formed by Na⁺, Mg²⁺, and Fe²⁺ Binding with Benzene Molecules. *J. Phys. Chem. A* **2014**, 118, 9500–9511.
- (12) Wang, H. Formation of Nascent Soot and Other Condensed-Phase Materials in Flames. *Proc. Combust. Inst.* **2011**, 33, 41–67.
- (13) Schuetz, C. A.; Frenklach, M. Nucleation of Soot: Molecular Dynamics Simulations of Pyrene Dimerization. *Proc. Combust. Inst.* **2002**, 29, 2307–2314.
- (14) Wong, D.; Whitesides, R.; Schuetz, C. A.; Frenklach, M. Molecular Dynamics Simulations of PAH Dimerization. In *Combustion Generated Fine Carbonaceous Particles: Proceedings of an International Workshop Held in Villa Orlandi*; Anacapri, May 13–16, 2007; Bockhorn, H., D'Anna, A., Sarofim, A. F., Wang, H., Eds.; KIT Scientific Publishing: Karlsruhe, Germany, 2009; pp 245–255.
- (15) Sabbah, H.; Biennier, L.; Klippenstein, S. J.; Sims, I. R.; Rowe, B. R. Exploring the Role of PAHs in the Formation of Soot: Pyrene Dimerization. *J. Phys. Chem. Lett.* **2010**, 1, 2962–2967.
- (16) Elvati, P.; Violi, A. Thermodynamics of Poly-Aromatic Hydrocarbon Clustering and the Effects of Substituted Aliphatic Chains. *Proc. Combust. Inst.* **2013**, 34, 1837–1843.
- (17) Brumbaugh, D. V.; Kenny, J. E.; Levy, D. H. Vibrational Predissociation and Intramolecular Vibrational Relaxation in Electronically Excited S-Tetrazine–Argon van der Waals Complex. *J. Chem. Phys.* **1983**, 78, 3415–3434.
- (18) Miller, R. E.; Vohralik, P. F.; Watts, R. O. Sub-Doppler Resolution Infrared Spectroscopy of the Acetylene Dimer: A Direct Measurement of the Predissociation Lifetime. *J. Chem. Phys.* **1984**, 80, 5453–5457.

- (19) Jucks, K. W.; Miller, R. E. Infrared Spectroscopy of the Hydrogen Cyanide Dimer. *J. Chem. Phys.* **1988**, *88*, 6059–6067.
- (20) Vande Linde, S. R.; Hase, W. L. Trajectory Studies of S_N2 Nucleophilic Substitution. I. Dynamics of $Cl^- + CH_3Cl$ Reactive Collisions. *J. Chem. Phys.* **1990**, *93*, 7962–7980.
- (21) Peslherbe, G. H.; Wang, H.; Hase, W. L. Unimolecular Dynamics of $Cl^- \cdots CH_3Cl$ Intermolecular Complexes Formed by $Cl^- + CH_3Cl$ Association. *J. Chem. Phys.* **1995**, *102*, 5626–5635.
- (22) Manikandan, P.; Zhang, J.; Hase, W. L. Chemical Dynamics Simulations of $X^- + CH_3Y \rightarrow XCH_3 + Y^-$ Gas-Phase S_N2 Nucleophilic Substitution Reactions. Nonstatistical Dynamics and Nontraditional Reaction Mechanisms. *J. Phys. Chem. A* **2012**, *116*, 3061–3080.
- (23) Baer, T.; Hase, W. L. *Unimolecular Reaction Dynamics. Theory and Experiments*; Oxford: New York, 1996.
- (24) Bunker, D. L.; Hase, W. L. On Non-RRKM Unimolecular Kinetics: Molecules in General and CH_3NC in Particular. *J. Chem. Phys.* **1973**, *59*, 4621–4632.
- (25) Hase, W. L. Unimolecular and Intramolecular Dynamics: Relationship to Potential Energy Surface Properties. *J. Phys. Chem.* **1986**, *90*, 365–374.
- (26) Lourderaj, U.; Hase, W. L. Theoretical and Computational Studies of Non-RRKM Unimolecular Dynamics. *J. Phys. Chem. A* **2009**, *113*, 2236–2253.
- (27) Rice, O. K. Several Remarks on the Energy Exchange within Molecules and Between Molecules during Collisions. *Z. Phys. Chem. B* **1930**, *7*, 226–233.
- (28) Jorgensen, W. L.; Severance, D. L. Aromatic–Aromatic Interactions: Free Energy Profiles for the Benzene Dimer in Water, Chloroform, and Liquid Benzene. *J. Am. Chem. Soc.* **1990**, *112*, 4768–4774.
- (29) Lee, E. C.; Kim, D.; Jurečka, P.; Tarakeshwar, P.; Hobza, P.; Kim, K. S. Understanding of Assembly Phenomena by Aromatic–Aromatic Interactions: Benzene Dimer and the Substituted Systems. *J. Phys. Chem. A* **2007**, *111*, 3446–3457.
- (30) Špirko, V.; Engkvist, O.; Soldán, P.; Selzle, H.; Schlag, E. W.; Hobza, P. Structure and Vibrational Dynamics of the Benzene Dimer. *J. Chem. Phys.* **1999**, *111*, 572–582.
- (31) Pulay, P.; Fogarasi, G.; Boggs, J. Force Field, Dipole Moment Derivatives, and Vibrational Constants of Benzene from a Combination of Experimental and *Ab Initio* Quantum Chemical Information. *J. Chem. Phys.* **1981**, *74*, 3999–4014.
- (32) Pulay, P. The Force Constants of Benzene: Local Many-Body Perturbation Theory vs New Experiment. *J. Chem. Phys.* **1986**, *85*, 1703–1704.
- (33) Guo, H.; Karplus, M. *Ab Initio* for the Planar Vibrations of Benzene. *J. Chem. Phys.* **1988**, *89*, 4235–4245.
- (34) Ozkabak, A. G.; Goodman, L.; Thakur, S. N.; Krogh-Jespersen, K. Complete Harmonic Force Field for Benzene Ground State In-Plane Vibrations. *J. Chem. Phys.* **1985**, *83*, 6047–6048. Ozkabak, A. G.; Goodman, L.; Thakur, S. N.; Krogh-Jespersen, K. Erratum: Complete Harmonic Force Field for Benzene Ground State In-Plane Vibrations. *J. Chem. Phys.* **1986**, *85*, 2346–2347.
- (35) Thakur, S.; Goodman, L.; Ozkabak, A. G. The Benzene Ground State Potential Surface. I. Fundamental Frequencies for the Planar Vibrations. *J. Chem. Phys.* **1986**, *84*, 6642–6656.
- (36) Ozkabak, A. G.; Goodman, L. The Benzene Ground State Potential Surface. II. Harmonic Force Field for the Planar Vibrations. *J. Chem. Phys.* **1987**, *87*, 5264–5214.
- (37) Goodman, L.; Ozkabak, A. G.; Thakur, S. N. A Benchmark Vibrational Potential Surface: Ground-State Benzene. *J. Phys. Chem.* **1991**, *95*, 9044–9058.
- (38) Hase, W. L.; Duchovic, R. J.; Hu, X.; Komornicki, A.; Lim, K. F.; Lu, D. H.; Peslherbe, G. H.; Swamy, S. R.; Vande Linde, S. R.; Varandas, A.; et al. VENUS96: A General Chemical Dynamics Computer Program. *Quantum Chemistry Program Exchange (QCPE) Bulletin* **1996**, *16*, 671.
- (39) Hu, X.; Hase, W. L.; Pirraglia, T. Vectorization of the General Monte Carlo Classical Trajectory Program VENUS. *J. Comput. Chem.* **1991**, *12*, 1014–1024.
- (40) McQuarrie, D. A. *Statistical Mechanics*; Harper & Row: New York, 1975.
- (41) Park, K.; Engelkemier, J.; Persico, M.; Manikandan, P.; Hase, W. L. Algorithms for Sampling a Quantum Microcanonical Ensemble of Harmonic Oscillators at Potential Minima and Conical Intersections. *J. Phys. Chem. A* **2011**, *115*, 6603–6609.
- (42) Peslherbe, G. H.; Hase, W. L. Statistical Anharmonic Unimolecular Rate Constants for the Dissociation of Fluxional Molecule. Application to Aluminum Clusters. *J. Chem. Phys.* **1996**, *105*, 7432–7447.
- (43) Manikandan, P.; Hase, W. L. Comparison of Classical Chemical Dynamics Simulations of the Unimolecular Decomposition of Classical and Quantum Microcanonical Ensembles. *J. Chem. Phys.* **2012**, *136*, 184110.
- (44) Lourderaj, U.; McAfee, J. L.; Hase, W. L. Potential Energy Surface and Unimolecular Dynamics of Stretched *n*-Butane. *J. Chem. Phys.* **2008**, *129*, 094701.
- (45) Kolakkandy, S.; Paul, A. K.; Pratihari, S.; Kohale, S. C.; Barnes, G. L.; Wang, H.; Hase, W. L. Energy and Temperature Dependent Dissociation of the $Na^+(benzene)_{1,2}$ Clusters: Importance of Anharmonicity. *J. Chem. Phys.* **2015**, *142*, 044306.
- (46) Hase, W. L.; Simons, J. W. Excitation Energies of Chemically Activated Isobutane and Neopentane and the Correlation of Their Decomposition Rates with Radical Recombination Rates. *J. Chem. Phys.* **1971**, *54*, 1277–1283.
- (47) Tsang, W. Evidence for Strongly Temperature-Dependent A Factors in Alkane Decomposition and High Heats of Formation for Alkyl Radicals. *Int. J. Chem. Kinet.* **1978**, *10*, 821–837.
- (48) Al-Alami, M. Z.; Kiefer, J. H. Shock-Tube Study of Propane Pyrolysis. Rate of Initial Dissociation from 1400 to 2300 K. *J. Phys. Chem.* **1983**, *87*, 499–506.
- (49) Oehlschlaeger, M. A.; Davidson, D. F.; Hanson, R. K. High-Temperature Thermal Decomposition of Isobutane and *n*-Butane behind Shock Waves. *J. Phys. Chem. A* **2004**, *108*, 4247–4253.
- (50) Vande Linde, S. R.; Mondro, S. L.; Hase, W. L. Transition States and Rate Constants for Ion–Molecule Association. II. $Li^+ + (CH_3)_2O \rightarrow Li^+[(CH_3)_2O]$. *J. Chem. Phys.* **1987**, *86*, 1348–1355.
- (51) Lu, D.-H.; Hase, W. L. Classical Trajectory Calculation of the Benzene Overtone Spectra. *J. Phys. Chem.* **1988**, *92*, 3217–3225.
- (52) Lu, D.-H.; Hase, W. L. Classical Mechanics of Intramolecular Vibrational Energy Flow in Benzene. V. Effect of Zero-Point Energy Motion. *J. Chem. Phys.* **1989**, *91*, 7490–7497.
- (53) Wyatt, R. E.; Iung, C.; Leforestier, C. Quantum Dynamics of Overtone Relaxation in Benzene. II. 16 Mode Model for Relaxation from $CH(v = 3)$. *J. Chem. Phys.* **1992**, *97*, 3477–3486.
- (54) Stock, G. Classical Simulation of Quantum Energy Flow in Biomolecules. *Phys. Rev. Lett.* **2009**, *102*, 118301.
- (55) Paul, A. K.; Kohale, S. C.; Pratihari, S.; Sun, R.; North, S. W.; Hase, W. L. A Unified Model for Simulating Liquid and Gas Phase, Intermolecular Energy Transfer: $N_2 + C_6F_6$ Collisions. *J. Chem. Phys.* **2014**, *140*, 194103.
- (56) Li, C.; Ross, P.; Szulejko, J. E.; McMahon, T. B. High-Pressure Mass Spectrometric Investigations of the Potential Energy Surfaces of Gas-Phase S_N2 Reactions. *J. Am. Chem. Soc.* **1996**, *118*, 9360–9367.
- (57) Wester, R.; Bragg, A. E.; Davis, A. V.; Neumark, D. M. Time-Resolved Study of the Symmetric S_N2 -Reaction $I^- + CH_3I$. *J. Chem. Phys.* **2003**, *119*, 10032–10039.
- (58) Mikosch, J.; Otto, R.; Trippel, S.; Eichhorn, C.; Weidemüller, M.; Wester, R. Inverse Temperature Dependent Lifetimes of Transient S_N2 Ion-Dipole Complexes. *J. Phys. Chem. A* **2008**, *112*, 10448–10452.
- (59) Lu, D.-h.; Hase, W. L. Classical Mechanics of Intramolecular Vibrational Energy Flow in Benzene. IV. Models with Reduced Dimensionality. *J. Chem. Phys.* **1988**, *89*, 6723–6736.
- (60) Miller, W. H.; Hase, W. L.; Darling, C. L. A Simple Model for Correcting the Zero Point Energy Problem in Classic Trajectory Simulations of Polyatomic Molecules. *J. Chem. Phys.* **1989**, *91*, 2863–2868.

(61) Peslherbe, G. H.; Hase, W. L. Analysis and Extension of a Model for Constraining Zero-Point Energy Flow in Classical Trajectory Simulations. *J. Chem. Phys.* **1994**, *100*, 1179–1189.

(62) Marcus, R. A.; Hase, W. L.; Swamy, K. N. RRKM and Non-RRKM Behavior in Chemical Activation Studies. *J. Phys. Chem.* **1984**, *88*, 6717–6720.

(63) Ezra, G. S.; Waalkens, H.; Wiggins, S. Microcanonical Rates, Gap Times, and Phase Space Dividing Surfaces. *J. Chem. Phys.* **2009**, *130*, 164118.